No.	Signature	Interpretation	Attack Labels	Individual Precision
1	Flow protocol: $\langle 17.0, 17.0\rangle$ ∧ Flow inter-arrival time standard deviation: $\langle 0.05, 0.05\rangle$ ∧ Forward inter-arrival time mean: $\langle 0.11, 0.11\rangle$ ∧ Maximum forward packet length: $\langle 77.0, 77.0\rangle$ ∧ Total number of forward packets: $\langle 5.0, 5.0\rangle$ ∧ Total number of backward packets: $\langle 0.0, 0.0\rangle$	Compact unidirectional UDP burst. The flow contains five forward packets, no backward packets, and a maximum forward packet length of 77 bytes, with short inter-arrival times. This one-way pattern is associated with action-stage traffic but does not reveal whether the payload carries a command or trigger. Operators can verify it using the destination port, payload, recurrence, and target response.	action	100%
2	Minimum flow inter-arrival time: $\langle 0.00027251244, 0.00041007996\rangle$ ∧ Minimum forward inter-arrival time: $\langle 0.00033426285, 0.00041246414\rangle$ ∧ Backward inter-arrival time standard deviation: $\langle -1.0, -1.0\rangle$ ∧ FIN flag: $\langle 0, 0\rangle$	Rapid traffic with limited reverse-direction evidence. Minimum inter-arrival times of approximately 0.27–0.41 ms indicate at least one rapid packet burst, while backward timing is unavailable and no FIN flag is observed. This pattern is shared by action and reconnaissance traffic rather than identifying a specific mechanism. Operators can distinguish them using the transport protocol, host or port fan-out, handshake state, and packet contents.	action, reconnaissance	100%
3	Flow protocol: $\langle 17.0, 17.0\rangle$ ∧ Flow inter-arrival time standard deviation: $\langle 0.01, 0.01\rangle$ ∧ Maximum forward packet length: $\langle 60.0, 60.0\rangle$ ∧ Minimum forward packet length: $\langle 60.0, 60.0\rangle$ ∧ Forward packets per second: $\langle 5.0, 5.0\rangle$ ∧ Total number of backward packets: $\langle 0.0, 0.0\rangle$	Repeated fixed-size unidirectional UDP traffic. Forward packets are 60 bytes and transmitted at five packets per second with low timing variation, while no backward packets are observed. This is consistent with automated action-stage traffic but does not alone identify command signaling or flooding. Operators can check payload similarity, repetition duration, aggregate rate, destination port, and source count.	action	100%
4	Flow protocol: $\langle 17.0, 17.0\rangle$ ∧ Flow packets per second: $\langle 1.0, 1.0\rangle$ ∧ Forward inter-arrival time standard deviation: $\langle 0.0, 0.0\rangle$ ∧ Maximum forward packet length: $\langle 77.0, 77.0\rangle$ ∧ Total number of forward packets: $\langle 1.0, 1.0\rangle$ ∧ Total number of backward packets: $\langle 0.0, 0.0\rangle$	Single unidirectional UDP datagram. The flow contains one 77-byte forward packet and no backward packet. Its zero inter-arrival-time variation is expected from a single-packet flow and is not evidence of regular timing. Operators should inspect the destination port, payload, recurrence, preceding reconnaissance, and subsequent target response.	action	100%

Table 10: Four representative signatures from `Mirai`. Those signatures are used to detect denial-of-service attacks, which accounts for most of the attacks in the dataset.